

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM)

Sem - I University Examination, 2010-11

ME – 702 Advances in Manufacturing Technology

Date: 04.01.2011, Tuesday

Time: 10:30 a.m. to 01:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

- Q1 (a)** What are the defects that are generally found in welding? Describe their causes and remedies **03**
- (b)** Is it possible to use centre lathe for friction welding? Justify your answer **03**
- (c)** Explain the basic characteristics of welding power sources. **05**
- Q2 (a)** Discuss frequency and safety regulations and their impact on RFID systems. **06**
- (b)** Discuss the classes, types and designs of readers for tag interrogation. **06**
- OR**
- Q2 (a)** Explain the phenomenon of Sputtering in PVD. **06**
- (b)** Explain with example the difference between CVD and PVD **06**
- Q3 (a)** Explain the components of Simulation. **04**
- (b)** What is the role of modeling and simulation in industrial product design and development? **04**
- (c)** Enlist simulation software for manufacturing applications. Explain GPSS/H in brief. **04**
- OR**
- Q3** **Write Short Notes (Any Three)** **12**
- (a)** Weld Distortion and Warpage
- (b)** High Frequency Resistance Welding
- (c)** Welding Automation and Robotics
- (d)** Micro-EDM

SECTION - II

- Q4 (a)** Explain with principle Laser Beam Machining stating advantages, limitations & applications 05
- (b)** Given the following data 06

X	6	2	10	4	8
Y	9	11	5	8	7

Find the two regression equations (S_x & S_y) and calculate the standard error of estimate.

- Q5 (a)** Enlist the steps for evaluation for manufacturability in manufacturing process model. 02
- (b)** List the characteristics of Hypothesis & differentiate between Null Hypothesis & Alternate Hypothesis. 04
- (c)** Describe the EDM process in detail giving important process parameters, working principle, advantages, limitations & field applications. 06

OR

- Q5 (a)** Sketch an outline of Manufacturing process model in Virtual Manufacturing. 03
- (b)** Explain one tailed & two tailed tests for hypothesis testing. 04
- (c)** Explain the principle of ECG. List the advantages of ECG over conventional grinding and its product applications. 05

- Q6 (a)** Explain the Abrasive Jet Machining with the help of a schematic diagram. Name the abrasives and carrier gases used in AJM. 04
- (b)** Set up ANOVA table for the following information relating to three drugs testing to judge effectiveness in reducing blood pressure for three different groups of people. 08

		Drug		
		X	Y	Z
Group	A	6	10	11
		10	9	11
	B	18	14	10
		17	12	11
	C	10	11	8
		11	11	7

Do the drugs act differently? Are the different groups of people affected differently? Is the interaction term significant? (Test at 5% significance level)
(Given: $F(2,9) = 4.26$; $F(4,9) = 3.63$)

OR

- Q6 Write short notes (Any Three)**

12

- (a)** Explosive forming
- (b)** Approaches for Experimentation
- (c)** Tube & Sheet Hydroforming
- (d)** CO₂ LASER